

Guided Reflection in AI-Assisted Decision-Making: Effects on AI Overreliance and Decision Accuracy

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Abstract

People often rely on heuristic reasoning when receiving algorithm advice, and this reliance leads to biased decisions that undermine the effectiveness of human-AI collaboration. Such bias persists even when individuals are given more time to deliberate or provided more information about AI, as they may lack the awareness or ability to engage in systematic reasoning. In this paper, we explore how guided reflection may enhance decision-making performance in human-AI collaboration by prompting a systematic reasoning process. We conducted an experiment with 178 participants, comparing decision-making behavior across three conditions: AI, explainable AI (XAI), and XAI with reflection. The results demonstrate that reflection significantly reduced over-reliance on AI and improved decision accuracy. Individuals with a high need for cognition and a high perceived understanding of AI benefited more from reflection. Furthermore, our study uncovers distinct patterns of cognitive processing and belief adjustment across different experimental conditions. Our findings provide a practical strategy for fostering cognitive engagement and contribute to a deeper understanding of human cognitive processes in AI-assisted decision-making.

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CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI; Human computer interaction (HCI); Interaction paradigms; Collaborative interaction.

Keywords

Reflection, Explainable AI, AI-assisted decision-making, Cognitive process

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1 Introduction

AI systems are increasingly integrated into diverse domains to support human decision-making, including criminal justice [35], financial investment [27, 48], hiring [42], and medical diagnosis [21]. However, this paradigm of AI-assisted decision-making has not produced better outcomes than human judgment or AI alone [34]. Empirical evidence suggests that users often exhibit either distrust in AI systems [43] or over-reliance on their outputs [28], leading to inappropriate utilization of algorithmic advice. To address this issue, researchers have advocated for algorithmic transparency as a means of fostering trust [50]. Yet, prior studies show that collaboration with explainable AI does not necessarily improve performance [4]. Human biases persist [4], and the explanations may create an illusion of understanding [39], which can further exacerbate over-reliance on AI.

These phenomena can be attributed to humans' tendency to rely on heuristic reasoning, also known as Type 1 processing [41], as the default cognitive process [21] when deciding whether to accept or reject algorithm advice. In AI-supported contexts, people's inherent biases toward or against AI may cause them to accept unbeneficial recommendations or ignore beneficial ones [21]. In XAI contexts, explanations often prompt individuals to form intuitive impressions of algorithmic competence, rather than analytically evaluating whether the explanations are reasonable [7]. This intuitive reasoning leads to greater over-reliance on XAI compared to the AI-supported context. Thus, promoting systematic reasoning (known as Type 2 processing) becomes crucial for mitigating these issues. Prior research has introduced a variety of interventions, drawing on ideas such as cognitive forcing functions [7], the Socratic questioning method [12], and deliberation theory [31], to encourage more analytical engagement with AI. However, this body of work largely overlooks a foundational theoretical issue: what determines whether individuals actually initiate and sustain Type 2 processing? Without addressing this question, intervention effects remain inconsistent, and several studies [2, 7] indicate that heuristic reasoning continues to dominate despite these efforts.

Dual-process theorists have extensively examined the characteristics of Type 1 and Type 2 processing [46]. Based on this foundation, a refined three-stage dual-process model [41] has been proposed to identify conflict detection and cognitive decoupling as the early and late mechanisms of Type 2 processing, respectively. According to the refined model, individuals first generate an intuitive response via Type 1 processing; this initial response may reflect biases, such as the intuitive impressions people form when encountering AI recommendations. In the second stage, individuals monitor for conflicts between Type 1 outputs—for example, by comparing AI-generated and human-generated information to detect inconsistencies. If no conflict is detected, the initial response carries forward to the third stage. Only when conflict is successfully detected do individuals engage Type 2 processing. In the third stage, Type 2 processing involves rationalization and decoupling. Only decoupling allows individuals to override intuitive responses and correct biased judgments. Thus, biases arise when conflict detection or cognitive decoupling fails.

Existing interventions aimed at promoting Type 2 processing in AI-assisted decision-making can be categorized into three types: interaction-level designs (e.g., delaying exposure to AI advice [7, 40], explanation-design interventions (e.g., questioning-style explanations [12]), and system-level designs (e.g., deliberative AI [31]). While each approach has shown some effectiveness, theoretical analysis [41] suggests their limitations in fully supporting Type 2 reasoning. Interaction-level interventions [7, 40], for instance, often do not explicitly guide conflict monitoring, which may lead to failures in conflict detection and a lack of engagement in Type 2 processing. Meanwhile, although interventions such as questioning explanations [12] and deliberative AI [31] can enhance conflict detection, they do not systematically encourage cognitive decoupling. Therefore, this study argues that to effectively promote Type 2 reasoning in AI-assisted decision-making, interventions must systematically address both conflict detection and cognitive decoupling as two core components.

To address the above limitations, this study draws on a refined three-stage dual-process model [41] and centers on conflict detection and cognitive decoupling as key mechanisms to propose and design guided reflection—a new intervention intended to ensure individuals engage in Type2 processing when collaborating with AI. Reflection is a cognitive activity through which individuals monitor, evaluate, and adjust their thinking [45], and it has proven effective in multiple decision-making contexts, including improving medical diagnostic accuracy [19, 37], enhancing information-seeking behaviors [45], strengthening online misinformation detection [22, 49], and increasing workplace performance [1]. Drawing on this foundation, we develop a guided reflection practice that encourages individuals to identify potential conflicts in the information they receive from both AI and human sources, override their initial intuitive responses to consider alternative approaches, and avoid focusing on justifying their initial judgments. The intervention consists of four reflection questions designed to promote conflict detection and cognitive decoupling. Specifically, the first question prompts conflict detection; the second and third questions encourage decoupling from intuitive responses; and the fourth question guides individuals to articulate the reasoning behind their updated decision before finalizing it. Through this reflective process, individuals integrate insights from both AI and human sources, update their personal beliefs accordingly, and ultimately make better-informed decisions based on the task at hand.

We investigate how this reflection practice influences human and AI collaboration from both outcome and process perspectives. Specifically, we examine its effects on performance outcomes, the boundary conditions under which these effects occur, and the ways in which reflection shapes belief change during decision-making. In addition, we examine how individuals reason when they are guided to detect conflicts and engage in cognitive decoupling, and how the reflection influences their overall experience of interacting with AI. Accordingly, we address the following five research questions: (1) How does reflection affect AI over-reliance and decision accuracy? (2) Are the effects of reflection moderated by individual characteristics? (3) How does reflection influence cognitive processes in AI-assisted decision making? (4) What types of conflicts do people identify, and what strategies and patterns do they adopt in final decision-making? (5) How does reflection shape user experiences?

To answer these questions, we conducted an experimental study using a two-stage diabetes prediction task. A total of 178 participants were recruited from Credamo (<https://www.credamo.cc>), a professional data collection platform similar to Mechanical Turk. In stage I, participants made an initial judgment independently; in stage II, they made final decisions with the assistance of either an opaque AI (AI), an explainable AI (XAI), or an explainable AI with the reflection (XAI+Reflection) intervention. We compared AI over-reliance and decision accuracy across these conditions. The results show that reflection significantly reduced over-reliance and improved decision accuracy. Further analyses revealed that individuals with a high need for cognition benefited more from reflection, consistent with prior findings [1]. Moreover, we found that participants with a high perceived understanding of AI in the XAI+Reflection condition achieved significantly greater accuracy and demonstrated less over-reliance than those in the XAI condition. By examining participants' belief changes from stage I to

stage II, we further shed light on how reflection shaped cognitive processes in AI-assisted decision-making. Additionally, we compared user experience across conditions to more comprehensively illustrate the broader impact of reflection on users.

In summary, this study makes three contributions. First, we introduce reflection as an interaction design intervention in human–AI decision-making, centered on conflict detection and cognitive decoupling, and demonstrate its potential to reduce over-reliance on AI while improving decision accuracy. This underscores the importance of moving beyond purely technical solutions, such as transparency [12] or uncertainty visualization [9], toward cognitive-level interventions that actively engage users in more systematic reasoning.

Second, we examine the boundary conditions of reflection by considering the roles of individual differences, specifically the need for cognition and the perceived understanding of AI. Extending prior research [7], we find that decision makers who reflected on both their own thoughts and algorithmic advice benefited more when they were capable of processing AI-generated information.

Third, we investigate AI over-reliance and human decision-making through the lens of changes in individuals' beliefs, providing a more fine-grained understanding of the mechanisms underlying human behavior in AI-assisted decision-making. Our findings reveal that reflection encouraged participants to recognize both inaccuracies in AI recommendations and the biased beliefs underlying their own decision-making, thereby demonstrating greater engagement in systematic reasoning.

2 Related work

2.1 Cognitive challenges in AI-assisted decision making

The paradigm of AI-assisted decision making refers to collaborative settings in which humans make decisions with the support of AI systems [3, 7, 31], where the goal is to achieve a performance that exceeds what either humans or AI could accomplish alone. Despite the potential, the paradigm faces two major cognitive challenges that limit its effectiveness.

The first cognitive challenge lies in people's inaccurate perceptions of AI capabilities, which manifest as both underestimation [34, 43] and overestimation [28]. On the one hand, prior research has suggested that demonstrating AI's ability to learn from mistakes [44] can increase human trust and encourage the adoption of AI recommendations. On the other hand, studies have proposed explanations [8, 12], performance indicators [9], and confidence levels [7, 32] as ways to help users better understand AI's strengths and limitations, thereby fostering appropriate reliance. However, establishing appropriate reliance is not sufficient. To fully leverage the complementary strengths of humans and AI [2], decision makers must go beyond forming accurate perceptions of AI capabilities and actively learn from AI advice. Yet, relatively little research has explored strategies that explicitly guide people in combining human and AI wisdom in decision-making [31]. Our study addresses this gap by proposing reflection as a strategy to facilitate not only appropriate reliance on AI but also the effective utilization of both human and AI domain knowledge.

The second cognitive challenge concerns the dominance of heuristic reasoning when people interact with AI [7, 29], which is closely related to the first challenge and may partly account for its occurrence. To counteract this tendency, prior studies have proposed various system- and interaction-level interventions aimed at encouraging more analytic thinking. For example, in terms of system design, Ma et al. [31] introduced a human–AI deliberation framework, which enables users to discuss conflicting opinions with AI before revising their own judgments. Danry et al. [12] proposed framing explanations as questions, which significantly improved logical discernment accuracy. In terms of interaction design, interventions such as presenting AI uncertainty [9] have been shown to foster more appropriate reliance. Zana et al. [7] proposed three cognitive forcing interventions to compel people to engage in analytic thinking, including selectively displaying AI recommendations based on demand, asking individuals to make initial decisions before showing the AI's suggestions, or requiring users to wait for 30 seconds before viewing AI outputs. However, these methods have produced mixed results [2, 7]. Some strategies primarily address the first challenge by helping users calibrate their perceptions of AI, rather than directly promoting systematic reasoning. This leaves open the question of how to effectively guide individuals to move beyond heuristics and engage in deeper cognitive processing.

Two closely related works are those by Ma et al. [31, 33] who proposed a deliberative AI [31] designed to prompt users to identify conflicts between human judgment and AI output, as well as a human self-confidence calibration mechanism [33] that encourages individuals to consider opposing perspectives. These approaches make valuable use of conflict detection to stimulate Type 2 thinking. At the same time, they place comparatively less emphasis on the role of cognitive decoupling in supporting more systematic forms of reasoning. According to the three-stage dual-process model [41], the absence of explicit support for rationalization and decoupling may limit the extent to which users can fully engage with detected conflicts, which in turn may influence the effectiveness of human–AI collaboration.

Another relevant contribution is Zana et al.'s study [8], which introduced a contrastive explanation framework to clarify differences between AI decisions and human reasoning and demonstrated its effectiveness in improving human decision-making skills. Whereas this framework primarily facilitates learning through the passive reception of explanations, our study focuses on guiding individuals to actively reflect on and learn from both the AI and their own reasoning. Such active engagement allows for a more fine-grained examination of cognitive processes in human–AI collaboration, including individuals' capacities for conflict detection and cognitive decoupling. In addition, Zana et al.'s findings suggest that contrastive explanations may increase users' susceptibility to AI persuasion [8], which could make it more challenging to mitigate overreliance on AI.

Given that people may lack both awareness of the need for systematic reasoning [15] and the ability to sustain it [45], we propose guided reflection, centered on conflict detection and cognitive decoupling, as an intervention to explicitly foster analytic thinking in AI-assisted decision making.

2.2 Reflection in decision-making processes

Reflection is a cognitive process through which individuals monitor, evaluate, and adjust their thinking [45], thereby learning from their own reasoning. It involves the process of absorbing and evaluating new concepts within an individual's existing knowledge frameworks, connecting these concepts with the person's prior knowledge and experiences [18]. As a strategy for fostering deep thinking and learning, reflection has proven effective in multiple decision-making contexts, including medical diagnosis [19, 37], information seeking [45], online deliberation [49, 51], and employee development [1]. For instance, research in clinical decision-making has shown that encouraging physicians to reflect on their initial diagnoses can significantly improve diagnostic accuracy in complex cases [19, 36]. In the domain of information seeking, Saito and Miwa [45] designed a system that supported participants in reflecting on their search process, which in turn enhanced task performance. Similarly, in educational contexts, reflection prompts have been used to enhance learning and problem-solving skills [10].

The theoretical foundation of reflection lies in dual-process models [46], which distinguish between two modes of cognitive processing: Type 1, characterized by fast, automatic, and intuitive reasoning, and Type 2, which involves controlled, analytical, and effortful reasoning. Reflection involves deliberate, effortful processing that leads to deeper cognitive elaboration and behavioral change [41]. This theoretical framework supports the idea that reflection enhances learning and decision-making by fostering systematic engagement with experiences and reasoning [1].

In practice, reflection is often facilitated through prompts that guide individuals to articulate their thoughts [36, 37, 51], evaluate their experiences [1, 45], or consider alternative perspectives [19, 49]. For example, in medical diagnosis, participants were asked to explicitly record findings that supported and contradicted their initial diagnosis [36, 37]. Participants in a work-simulation task [1] were instructed to reflect by writing down what they had done correctly and where things had gone wrong. Consistent with this line of research, our study also employs structured reflection guidance. To encourage systematic reasoning in AI-assisted decision-making, participants were asked to complete a three-step reflection process: (1) detect conflicts between their own judgment and the AI's recommendation, (2) consider both confirmatory and contradictory perspectives, and (3) articulate the reasons for their final decision.

Overall, reflection has been increasingly applied in the field of Human–Computer Interaction (HCI) as a means of improving decision-making and enhancing critical thinking [22, 25, 49, 51]. Given the growing prevalence of AI-assisted decision-making systems, reflection represents a promising tool for strengthening human–AI collaboration. It offers individuals an opportunity to evaluate information from both their own judgment and AI recommendations, to reflect on conflicts and to engage in deeper cognitive elaboration, which can potentially address two cognitive challenges. However, existing research has rarely examined its role in addressing the two specific cognitive challenges of AI-assisted contexts—namely, the tendency to overrely on algorithmic advice and the dominance of heuristic reasoning. This gap underscores the need to investigate how reflection can be effectively integrated into AI-assisted systems to foster systematic reasoning, mitigate

over-reliance, and ultimately improve human–AI decision-making outcomes.

3 HYPOTHESIS DEVELOPMENT

Drawing on the three-stage dual process model [41], this study proposes a reflection intervention that incorporates conflict detection and cognitive decoupling in order to guide individuals to engage in Type 2 processing when collaborating with XAI. The purpose of eliciting Type 2 processing is to address the persistent problem of over-reliance in XAI contexts. Research Questions 1 and 2 therefore focus on how the combination of XAI and reflection influences the performance of human and AI collaboration, as well as the boundary conditions of this influence. Based on these research questions, we develop two sets of hypotheses.

Reflection is a purposeful and multi-stage cognitive activity [45] through which individuals systematically process both their own thoughts and the reasoning provided by AI. When individuals engage in reflection, they critically examine the arguments underlying AI recommendations. This promotes a more objective understanding of algorithmic advice and a more accurate assessment of the AI system's competence [21]. As a result, individuals are expected to calibrate their trust in AI and reduce over-reliance, which may lead to fewer instances of accepting incorrect AI recommendations in their final decisions [31, 33]. Furthermore, reflection helps individuals recognize their own limitations. By examining the rationales provided by AI, individuals acquire additional task-relevant information that can supplement gaps in their own knowledge. This enriched understanding enables them to make more informed judgments, which may ultimately increase decision accuracy. Therefore, we hypothesize:

H1a. XAI+reflection will reduce over-reliance on AI compared with XAI.

H1b. XAI+reflection will improve decision accuracy compared with XAI.

Need for cognition refers to individuals' tendency to engage in and enjoy effortful cognitive activity [26]. Individuals high in need for cognition tend to engage in deeper thinking and perform cognitively challenging tasks, whereas individuals low in need for cognition prefer automatic processes [50]. Prior research has found that the effects of cognitive forcing functions in promoting analytic thinking depend on individuals' levels of need for cognition [7]. We expect that the reflection intervention will be more effective for participants high in need for cognition.

Perceived understanding of AI captures an individual's mental model of AI systems [14], that is, the user's knowledge of the system's components and how they interact. Participants who develop an accurate mental model of AI should be able to identify the accuracy and usefulness of information provided by AI—that is, to discern which information is valuable. By contrast, participants with low perceived understanding of AI may struggle to effectively utilize AI knowledge and instead merely justify their own initial thoughts. Prior research in finance [11] has demonstrated that traders who can evaluate the precision of signals benefit more from cognitive reflection, which involves actively updating beliefs based on signals from market orders. We expect that the reflection

intervention will be more effective for participants high in perceived understanding of AI. Thus, we hypothesize:

H2a. Individuals with a high need for cognition will benefit more from reflection than individuals with a low need for cognition.

H2b. Individuals with a high perceived understanding of AI will benefit more from reflection than those with a low perceived understanding of AI.

4 Experiment

We conducted a between-subjects experiment with three conditions: an opaque AI baseline (AI), an explainable AI without reflection (XAI), and an explainable AI with reflection intervention (XAI+Reflection). A total of 178 participants were recruited from Credamo to complete a two-stage diabetes prediction task. In Stage I, all participants made initial judgments independently; in Stage II, participants made final judgments under their assigned support condition. Credamo, which maintains a pool of three million registered subjects with diverse demographics, is one of the largest platforms for conducting behavioral research studies [20]. The quality of the platform has been demonstrated by publications in top-tier journals, including PNAS [20] and POM [52]. The study had five objectives: First, we compared AI over-reliance and decision accuracy across the three conditions to assess the effect of reflection. Second, we examined whether the effects of reflection varied with individual characteristics. Third, we compared belief changes from Stage I to Stage II to identify differences in the underlying decision-making processes across conditions. Fourth, we coded participants' reflections to distill key themes in their reflective thinking. Fifth, we compared user experience measures across conditions to assess how reflection influenced participants' experiences.

4.1 Task, dataset, and AI model

We selected diabetes prediction as the task context for three reasons. First, AI-assisted decision making in the medical domain has attracted substantial scholarly attention [16, 21], and reflection before decision-making is particularly important in medical contexts compared to many other domains. Prior research has also shown that the use of AI systems in diagnostic contexts can induce reflection [16]. Second, compared with other medical conditions, diabetes diagnosis is relatively familiar to the general public. Third, existing AI tools target diabetes screening or self-assessment [5, 53].

The task was based on the Pima Indians Diabetes dataset from Kaggle [53], which contains diagnostic information from 769 individuals. The dataset includes eight predictor variables: number of pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, and age. In addition, it contains a binary outcome variable indicating whether the individual has diabetes, with 0 representing no diabetes and 1 representing diabetes.

The dataset was randomly divided into two parts: 70% was used as the training set for model development, and the remaining 30% served as the test set from which the experimental cases were selected. We employed a Random Forest model because it demonstrated the best performance compared with the other two algorithms tested (AdaBoost and XGBoost). The Random Forest model was trained using all available features and achieved a precision of 81% and a recall of 87%. Consequently, the predictions generated

by the Random Forest model on the test set were used as the AI suggestions presented to participants.

Four cases from the test set were selected for the experimental task. This decision was based on two considerations. First, feedback from a pilot study indicated that four cases provided an appropriate level of task complexity, given that participant was required to articulate their reasoning and make prediction for each case (the detailed experimental procedure is described in section 4.3). Second, prior studies [4, 31] with similar experimental designs have also employed four cases, suggesting that this number is both methodologically suitable and unlikely to compromise the validity of experimental outcomes. To further reduce task complexity, only five features were shown to participants when making decisions: fasting plasma glucose (mg/dL), insulin (μ U/mL), BMI (kg/m^2), diabetes pedigree function, and age. All participants viewed the same four cases, which were presented in a random order; the cases are shown in Figure 1.

To provide explanations for the model's predictions, we used SHAP (Shapley Additive exPlanations), a state-of-the-art XAI technique [30]. SHAP produces explanations by assigning additive contributions of individual features to the overall black-box model prediction. In the experiment, SHAP explanations were presented as bar plots: features pushing the prediction toward "has diabetes" were colored red, while those pushing toward "does not have diabetes" were colored green. For example, the middle panel on the right side of Figure 1 displays a SHAP bar plot in which the top bar (in green) is labeled with a value of -0.09. This value indicates that the feature "fasting plasma glucose" being equal to 90 mg/dL has a SHAP value of -0.09, meaning that this feature contributes negatively to the predicted probability of diabetes.

4.2 Conditions

AI condition. After making initial decisions in Stage I, participants were presented only with the AI's recommendation in Stage II before submitting their final decision and reporting beliefs about the relationship between the five indicators of the individuals and the diabetes prediction.

XAI condition. After making initial decisions in Stage I, participants were shown the AI's recommendation together with explanation plots generated by SHAP in Stage II. They then made their final decisions and reported their beliefs.

XAI+Reflection condition. After making initial decisions in Stage I, participants were shown the same AI's recommendation with explanation plots as in the XAI condition in Stage II. However, upon receiving this information, participants were guided to complete a three-step reflection. According to the three-stage dual-process model of analytic engagement proposed by Pennycook et al. [41], conflict detection and decoupling serve as early and late stages of analytic reasoning, respectively. This implies that without successful conflict detection, or if participants focus solely on justifying their initial response, they fail to engage in systematic reasoning. We designed a three-step reflection procedure (as shown in Figure 2). Step 1 (Q1) guided participants to identify potential conflicts; Step 2 (Q2–Q3) prompted them to decouple; and Step 3 (Q4) encouraged them to reconsider their final choice. Importantly, these steps were designed to be iterative, allowing participants to revisit

A

Medical indicators	Case1	Case2	Case3	Case4	Dataset range
Glucose(mg/dL)	81	90	104	152	44-200
Insulin(μ U/mL)	102	59	169	29	14-800
BMI	27.7	25.1	31	26.8	18.2-67.1
Diabetes predigree function	0.3	1.3	0.5	0.7	0.07-2.4
Age	25	25	38	43	21-81

Task 1/4

Q1. Please make decision on whether individual 1 has diabetes.

☐ Individual 1 has diabetes ☐ Individual 1 does not have diabetes

Q2. Please using the slider to elicit the thoughts: moving the slider to the right indicates that the feature increases the probability of diabetes, moving it to the left indicates that the feature decreases the probability of diabetes.

Fasting plasma glucose(mg/dL): 90

Insulin(μ U/mL): 59

BMI: 25.1

Diabetes predigree function: 1.3

Age: 25

Q3. Enter your confidence in your initial decision. Choose from 1 (very uncertain) to 7 (very certain).

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

Next Task

B

Task 1/4 Condition: AI

- AI's suggestion: individual 1 does not have diabetes
- Your initial decision: individual 1 has diabetes

Task 1/4 Condition: XAI

- AI's suggestion: individual 1 does not have diabetes
- Your initial decision: individual 1 has diabetes
- AI explanation:

Fasting plasma glucose(mg/dL): 90 -0.09

Insulin(μ U/mL): 59 -0.08

BMI: 25.1 -0.06

Diabetes predigree function: 1.3 0.08

Age: 25 -0.02

- A red bar indicates that the feature has a positive contribution, increasing the probability of a "diabetes" prediction.
- A green bar indicates that the feature has a negative contribution, decreasing the probability of a "diabetes" prediction.
- The number shown next to each bar represents the specific magnitude of that feature values' contribution to the model's prediction.
- For example, the data "-0.09" indicates that this feature decreases the model's predicted probability of diabetes by 9%.

Q1. Please make final decision on whether individual 1 has diabetes.

☐ Individual 1 has diabetes ☐ Individual 1 does not have diabetes

Q2. Please using the slider to elicit the thoughts: moving the slider to the right indicates that the feature increases the probability of diabetes, moving it to the left indicates that the feature decreases the probability of diabetes.

Fasting plasma glucose(mg/dL): 90

Insulin(μ U/mL): 59

BMI: 25.1

Diabetes predigree function: 1.3

Age: 25

Q3. Enter your confidence in your final decision. Choose from 1 (very uncertain) to 7 (very certain).

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

Figure 1: User interfaces for Stage I and for the AI and XAI conditions in Stage II. (A) The user interface for Stage I. The top section presents four example cases used in the experiment, each showing the medical indicator values for that case along with the corresponding value ranges in the dataset. The bottom section allows participants to make their initial decision and to use sliders to report the thoughts underlying that decision. (B) The user interface for Stage II in the AI condition and the XAI condition. The top section shows the AI recommendation presented in the AI condition. The middle section shows both the AI recommendation and the explanation presented in the XAI condition. The bottom section allows participants to make their final decision and to use sliders to report the thoughts underlying that decision.

and revise earlier reflections—for example, recognizing a conflict only after completing Step 2 and then returning to refine Step 1.

Specifically, in Step 1, participants reported whether they perceived any conflict between the AI's recommendations and explanations and their own initial thoughts. In Step 2, they wrote down what they had judged correctly and incorrectly during their initial decision-making, as well as which aspects of the AI's reasoning they considered accurate or flawed. In Step 3, participants chose whether to follow their initial decision or the AI's recommendation and provided written justification for their choice. After completing these reflection activities, participants made their final decisions and reported their beliefs.

4.3 Procedure

With institutional IRB approval, we conducted a between-subjects study. Participants first provided informed consent and were then randomly assigned to one of three conditions. Following consent, a brief introduction outlined the study's goals, tasks, and overall workflow. Participants then completed a short pre-questionnaire

to collect demographic information, AI-related background, and diabetes health literacy.

The main task consisted of two stages. In Stage I, participants reviewed four patient profiles (each with five features) and made initial diagnoses without AI assistance. For each case, they first recorded their binary diagnosis. They then reported their beliefs about the relationship between each feature and diabetes using sliders ranging from -1 to 1 , where positive values indicated that the feature supported a diabetes diagnosis and negative values indicated that the feature suggested the opposite. Finally, they rated their confidence in their initial diagnosis on a seven-point scale.

In Stage II, participants revisited the same four cases under their assigned condition. At the start of Stage II, participants were introduced to the machine learning method used in the experiment. Participants assigned to the condition including XAI additionally received training on interpreting SHAP explanations. The meaning of SHAP explanations was explained, after which participants completed a manipulation check question to ensure they understood

A Task 1/4

- AI's suggestion: individual 1 does not have diabetes
- Your initial decision: individual 1 has diabetes
- AI explanation:

Fasting plasma glucose(mg/dL): 90
-0.09

Insulin(μ U/mL): 59
-0.08

BMI: 25.1
-0.06

Diabetes predigree function: 1.3
0.08

Age: 25
-0.02

- A red bar indicates that the feature has a positive contribution, increasing the probability of a "diabetes" prediction.
- A green bar indicates that the feature has a negative contribution, decreasing the probability of a "diabetes" prediction.
- The number shown next to each bar represents the specific magnitude of that feature values' contribution to the model's prediction.
- For example, the data "-0.09" indicates that this feature decreases the model's predicted probability of diabetes by 9%.

Q5. Please make final decision on whether individual 1 has diabetes.

☐ Individual 1 has diabetes ☐ Individual 1 does not have diabetes

Q6. Please using the slider to elicit the thoughts: moving the slider to the right indicates that the feature increase the probability of diabetes, moving the slider to the left indicates that the feature decrease the probability of diabetes

Fasting plasma glucose(mg/dL): 90

Insulin(μ U/mL): 59

BMI: 25.1

Diabetes predigree function: 1.3

Age: 25

Q7. Enter your confidence in your final decision. Choose from 1 (very uncertain) to 7 (very certain).

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

B Condition: XAI+reflection

Three-step reflection procedure

Q1. Please reflect on whether the AI's suggestion and explanation conflict with your initial decision and thoughts in stage 1.

☐ No obvious conflict. The AI's suggestion and explanation are consistent with my initial decision and thoughts.

☐ Conflict exists. The AI's suggestion and explanation differ from my initial decision and thoughts.

Q2. Which parts of the AI's suggestion and explanation are reasonable, and which are unreasonable or surprising? Please give examples.

Q3. About your decision-making in stage 1, which of your thoughts were correct, and which need to be adjusted or were previously overlooked? Please give examples.

Q4. How would you make the decision again? (choose one of the two options)

☐ Stick to your initial decision. Please explain the reason below

☐ Revise initial decision based on the AI's suggestion and explanation. Please explain the reason below

Conflict detection

Decoupling: evaluating the correctness of both human and the AI's thoughts

Reflect on final decision

C

Figure 2: User interface for the XAI+Reflection condition in Stage II. (A) The AI's recommendation and explanation. (B) The three-step reflection procedure, consisting of four questions. (C) The final decision screen, where participants make their ultimate choice and use sliders to report the reasoning underlying that decision.

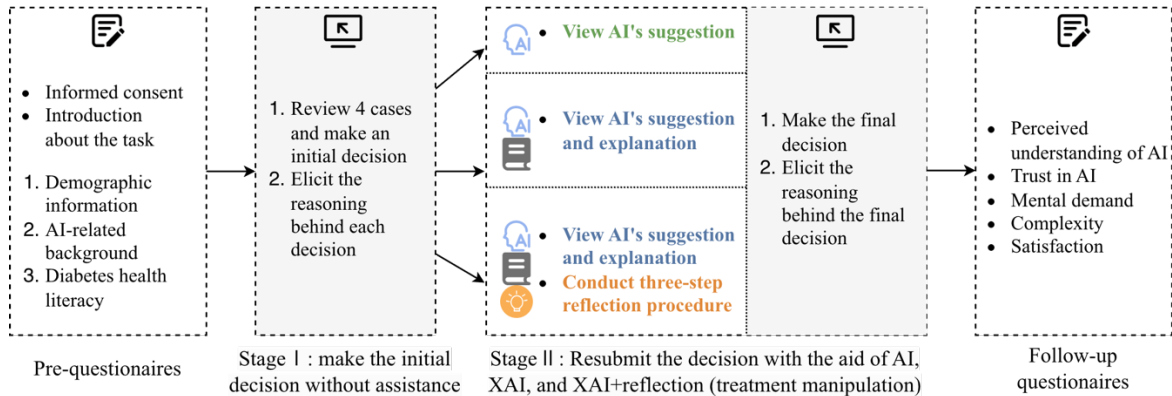


Figure 3: Overview of the experimental procedure.

the explanation. Participants who failed this check were excluded from the sample. Participants received the treatment corresponding to their assigned condition and then made final decisions and repeated the belief elicitation for the same four cases. Decisions and confidence ratings across both stages were recorded.

After completing all cases, participants filled out a post-questionnaire assessing subjective measures, including perceived

understanding of AI, trust in AI, mental demand, complexity, and satisfaction. The study concluded upon submission of the post-questionnaire.

4.4 Participants

We recruited 210 participants from the Credamo platform. To ensure data quality, participants were required to complete attention-check questions. After applying exclusion criteria and removing low-quality responses, 178 valid samples were retained for analysis. The final distribution across the three experimental conditions was as follows: AI ($n = 58$), XAI ($n = 59$), and XAI+Reflection ($n = 61$). Among the final sample, 56.2% self-identified as female ($n = 100$) and 43.8% as male ($n = 78$). The age distribution was relatively balanced across younger and mid-career groups: 69 participants aged 18–25, 75 aged 26–35, 24 aged 36–45, and 10 aged 45 or older. Regarding educational attainment, 30 participants held a master's or doctoral degree, 125 held a bachelor's degree, and 23 reported an education below the bachelor's level. In terms of prior experience with predictive software, 86.5% ($n = 154$) indicated familiarity with AI. Each participant completed the study in 10–20 minutes and received ¥10 as compensation.

4.5 Measurement

To answer the research questions, we collected both objective performance measures and subjective self-reported measures.

Decision accuracy. We calculated the accuracy of participants' final predictions relative to the ground truth [31].

Reliance. Over-reliance was measured as the percentage of cases where participants agreed with the AI when the AI made incorrect predictions, whereas under-reliance was measured as the percentage of cases where participants rejected the AI when the AI made correct predictions [7, 31].

Need for cognition. Need for cognition was measured using six scale items adapted from Gabriel et al. [26]. This scale contains statements such as “I would prefer complex to simple problems” and “I would rather do something that requires little thought than something that is sure to challenge my thinking abilities” (reverse scored).

Perceived understanding of AI. Perceived understanding of AI was measured using two items adapted from Wang and Yin [47].

Trust in AI. Trust in AI was measured using a single item assessing agreement with the statement: “I trust the AI model to provide reliable decision support” [7].

User experience. Mental demand and system complexity were both measured using items adapted from Zana et al. [7], and satisfaction was measured with one item adapted from Ma et al. [31].

All subjective measures were assessed using 7-point Likert scales.

4.6 Analysis

For the quantitative analysis, we first performed normality tests (Shapiro–Wilk) and found that the data did not fit the normality assumption. As a result, we adopted non-parametric tests for all subsequent analyses. Specifically, to compare decision accuracy, over-reliance, and user experience across the three experimental conditions, we employed Kruskal–Wallis tests with Bonferroni-adjusted post-hoc correction. All reported p -values were adjusted accordingly.

Additionally, we conducted an inductive thematic analysis [6, 23] to identify themes emerging from participants' reflections. Building on prior methodological guidance [13], the analysis proceeded in

two phases. The first phase focused on responses from Step 1 (Q1) and Step 2 (Q2–Q3) to examine whether participants detected conflicts during collaboration with AI and to identify the specific types of conflicts recognized. The second phase analyzed Step 3 (Q4) responses to extract strategies or patterns underlying participants' final decision-making. Two coders carried out the analysis of the reflective content. They first immersed themselves in the data and generated initial codes. Similar codes were then grouped to form potential themes. For example, codes such as “the AI's prediction is more accurate,” “the AI considers multiple indicators,” and “the AI has superior capabilities” all reflected participants' trust in the AI's competence, which led to the development of the theme “competence trust in AI”. The coders subsequently reviewed the themes to ensure coherence and distinctiveness, and then refined and named the themes following established guidelines [13]. This process was iterative, and the two coders discussed and resolved any coding discrepancies through consensus.

5 Results

5.1 How does reflection affect AI over-reliance and decision accuracy?

In this section, we present our findings in response to the first research question: (RQ1) How does reflection affect AI over-reliance and decision accuracy? Specifically, we examine the impact of reflection on AI over-reliance and decision accuracy.

5.1.1 AI Over-reliance. As shown in the right of Figure 4, participants in both the XAI+Reflection condition ($M = 0.34$, $SD = 0.06$) and the AI condition ($M = 0.33$, $SD = 0.06$) exhibited significantly less over-reliance on AI than those in the XAI condition ($M = 0.58$, $SD = 0.06$), with $p < 0.05$ for the comparison between the XAI+Reflection and XAI groups, and $p < 0.01$ for the comparison between the AI and XAI groups. Thus, H1(a) was supported. These results indicate that requiring participants to engage in reflection effectively mitigates blind agreement with AI predictions. This aligns with prior research [7, 31], which demonstrates that cognitive forcing functions and human-AI deliberation can reduce human over-reliance on AI.

We also compared under-reliance across the three conditions. The results in the left of Figure 4 showed no significant difference between the XAI+Reflection condition ($M = 0.05$, $SD = 0.17$) and the XAI condition ($M = 0.04$, $SD = 0.16$), indicating that reflection did not lead to under-reliance. The reflection intervention appeared to promote more appropriate reliance on AI. It encouraged individuals to think more critically about the AI's suggestions and actively assess whether the AI's predictions were correct or incorrect, fostering a more objective understanding of AI.

5.1.2 Decision accuracy. We first compared participants' initial decision accuracy across the three conditions. A non-parametric test showed no significant differences in Stage I decision accuracy across conditions ($p = 0.627$), indicating that our random assignment procedure was successful. We then examined decision accuracy in Stage II, which directly addresses our research question regarding the effect of reflection. The results showed that participants in the XAI+Reflection condition ($M = 0.881$, $SD = 0.020$) achieved significantly higher accuracy than those in both the XAI condition

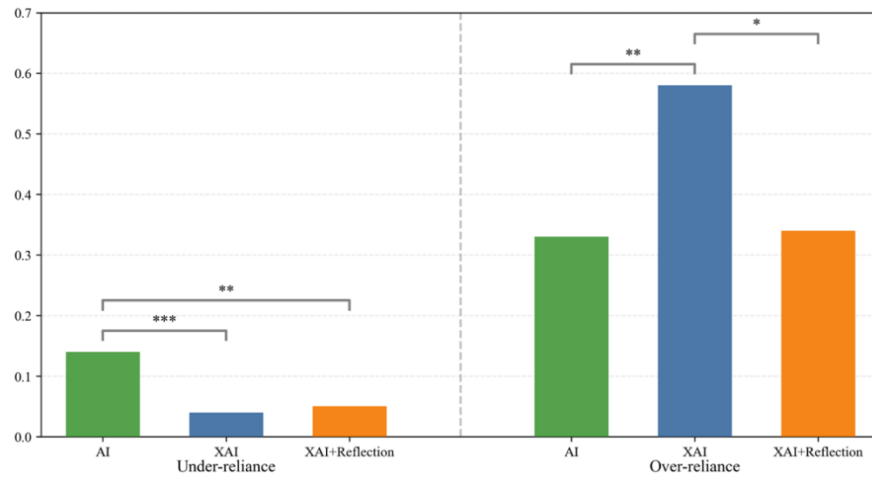


Figure 4: Under-reliance and over-reliance across different conditions. (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$)

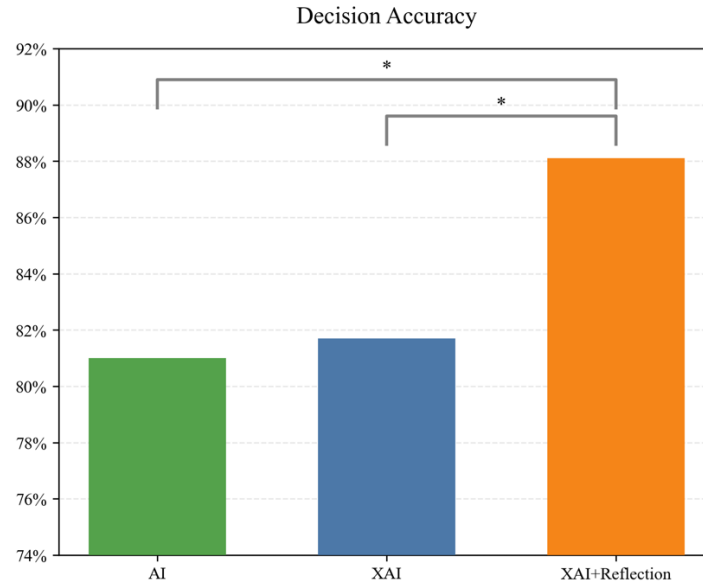


Figure 5: Decision accuracy in Stage II across different conditions. (*: $p < 0.05$)

($M = 0.817$, $SD = 0.018$, $p < 0.05$) and the AI condition ($M = 0.810$, $SD = 0.024$, $p < 0.05$). These findings support H1(b). We conducted an additional regression analysis (see Appendix) using the change in decision accuracy between the two stages as the dependent variable. After controlling for decision accuracy in Stage I and all other covariates potentially associated with the outcome, the XAI+Reflection condition remained significantly different from the XAI condition. This confirms the robustness of our main findings. This result suggests that incorporating a reflection into the XAI process can enhance human decision accuracy beyond both receiving explanations without reflection and relying solely on AI predictions. Reflection appears to encourage participants to scrutinize AI

outputs more carefully, which improves the quality of their final judgments.

5.2 What roles do individual characteristics play in influencing AI over-reliance and decision accuracy?

In this section, we present our findings addressing the second research question: (RQ2) Are the effects of reflection moderated by individual characteristics? Specifically, we examine how two individual factors—need for cognition and perceived understanding of AI—moderate the impact of reflection on AI over-reliance and decision accuracy.

Table 1: Decision accuracy and AI over-reliance by NFC group

	High NFC			Low NFC		
	Marginal means (standard errors)		Significance	Marginal means (standard errors)		Significance
	XAI+Reflection	XAI		XAI+Reflection	XAI	
Decision accuracy	0.90 (0.02)	0.83 (0.02)	$p=0.040<0.05$	0.85 (0.03)	0.80 (0.03)	$p=0.351, \text{ n.s.}$
Over-reliance	0.34 (0.08)	0.58 (0.09)	$p=0.040<0.05$	0.34 (0.08)	0.57 (0.09)	$p=0.195, \text{ n.s.}$

Table 2: Decision accuracy and AI over-reliance by perceived understanding of AI group

	High perceived understanding of AI			Low perceived understanding of AI		
	Marginal means (standard errors)		Significance	Marginal means (standard errors)		Significance
	XAI+Reflection	XAI		XAI+Reflection	XAI	
Decision accuracy	0.90 (0.02)	0.82 (0.02)	$p=0.022<0.05$	0.84 (0.04)	0.78 (0.04)	$p=0.202, \text{ n.s.}$
over-reliance	0.31 (0.07)	0.55 (0.07)	$p=0.028<0.05$	0.42 (0.11)	0.67 (0.12)	$p=0.147, \text{ n.s.}$

5.2.1 The moderating effect of need for cognition (NFC). Participants were divided into two groups, low NFC and high NFC, based on the median of the NFC score distribution. For high-NFC participants, the distributions of decision accuracy and over-reliance were significantly different between the XAI and the XAI+Reflection categories. High-NFC participants in the XAI+Reflection condition ($M = 0.90, SD = 0.02$) showed higher decision accuracy compared to those in the XAI condition ($M = 0.83, SD = 0.02, p < 0.05$). Additionally, high-NFC participants in the XAI+Reflection condition ($M = 0.34, SD = 0.08$) overrelied less than those in the XAI condition ($M = 0.58, SD = 0.09, p < 0.05$).

For participants with low NFC, there were no significant differences in decision accuracy and over-reliance between the XAI and XAI+Reflection conditions ($p > 0.05$). Thus, H2(a) was supported. The results suggest that need for cognition influences how reflection affects decision accuracy and over-reliance. Participants with high NFC benefit more from reflection, whereas reflection did not significantly enhance task performance for those with low NFC.

5.2.2 The moderating effect of perceived understanding of AI. Participants were divided into two groups, those with low perceived understanding of AI and those with high perceived understanding of AI, based on the median of the perceived understanding score distribution. For participants who reported a high perceived understanding of AI, the differences between XAI and XAI+Reflection were significant. Specifically, participants with high perceived understanding of AI in the XAI+Reflection condition achieved higher decision accuracy ($M = 0.90, SD = 0.02$) than those in the XAI condition ($M = 0.82, SD = 0.02, p < 0.05$). They also exhibited lower over-reliance in the XAI+Reflection condition ($M = 0.31, SD = 0.07$) compared to the XAI condition ($M = 0.55, SD = 0.07, p < 0.05$).

By contrast, for participants with low perceived understanding of AI, there were no significant differences in either decision accuracy or over-reliance between the XAI and XAI+Reflection conditions ($p > 0.05$). Thus, H2(b) was supported. These findings suggest

that the benefits of reflection are contingent on participants' self-perceived understanding of AI: reflection is more effective for those who feel more confident in their comprehension of AI concepts, while it provides limited advantages for those who perceive their understanding as low.

5.3 Differences in the decision-making process across conditions

To address Research Question 3, we examine whether the decision-making processes differ among participants across the three conditions. We analyze participants' belief adjustments (in terms of each feature's contribution to diagnosis) from Stage I to Stage II. First, we compare participants' absolute belief adjustments in the AI, XAI, and XAI+Reflection conditions. Second, we compare the direction of belief changes for participants in the AI, XAI, and XAI+Reflection conditions.

As shown in Figure 6, the average absolute belief adjustment for participants in the XAI condition ($M = 0.36, SD = 0.011$) is significantly higher than that in the XAI+Reflection condition ($M = 0.26, SD = 0.008, p < 0.001$) and the AI condition ($M = 0.17, SD = 0.006, p < 0.001$). The results indicate that participants in the XAI condition significantly changed their beliefs, whereas participants in the AI condition adjusted their beliefs to a lesser extent.

Furthermore, we analyze belief adjustments more explicitly by comparing the direction of belief changes for participants in three different conditions. The variables $Prior_{ijk}$ and $Post_{ijk}$, respectively, denote participant i 's prior and posterior beliefs about feature j 's contribution to the predicted probability that individual k has diabetes. Here, j denotes the feature index, with $j \in \{1, 2, 3, 4, 5\}$, and k denotes the individual index, with $k \in \{1, 2, 3, 4\}$. The $SHAP_{jk}$ denote the SHAP value for feature j , indicating its contribution to the predicted probability that individual k has diabetes.

Additionally, we define an indicator variable $align_with_SHAP_{ijk}$ as follows: $align_with_SHAP_{ijk} = 1$, if

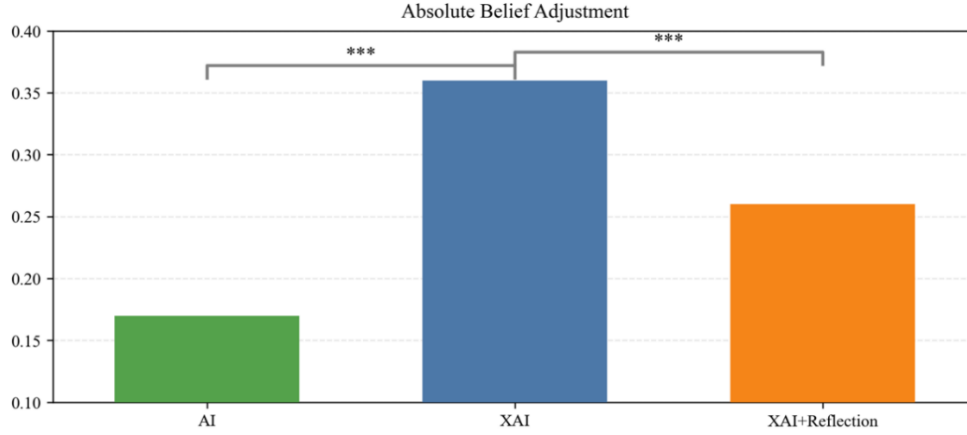


Figure 6: Absolute belief adjustment across different conditions

the sign of $Post_{ijk}$ matches the sign of $SHAP_{jk}$ (i.e., both are either positive or negative), and $align_with_SHAP_{ijk} = 0$, otherwise, when the signs do not match.

Thus, $align_with_SHAP_{ijk}$ captures whether participant i 's posterior belief about feature j aligns in sign with the corresponding SHAP value. Based on $align_with_SHAP_{ijk}$, we further define $n_align_with_SHAP_{ik} = \sum_{j=1}^5 align_with_SHAP_{ijk}$, which represents the total number of features for which participant i 's posterior beliefs align in sign with the SHAP values on whether individual k has diabetes.

Figure 7 shows the distribution of $n_align_with_SHAP_{ik}$ for participants in the XAI, and XAI+Reflection conditions. The distribution of $n_align_with_SHAP_{ik}$ for participants in the XAI condition is highly concentrated at 5, indicating that participants in this group mainly followed the SHAP values across all five features. The distribution for the XAI+Reflection condition is more dispersed, with peaks around $n_align_with_SHAP_{ik} = 3, 4$, and 5, suggesting that participants selectively followed the SHAP. There is a significant difference in the proportion of participants with $n_align_with_SHAP_{ik} = 5$ between the XAI ($M = 0.68, SD = 0.03$) and XAI+Reflection conditions ($M = 0.51, SD = 0.03, p < 0.001$). The results indicate that participants in the XAI condition tended to blindly follow the explanations without engaging in additional deliberation. In contrast, participants in the XAI+Reflection condition were more reflective and maintained prior beliefs for certain features.

Furthermore, we define an indicator variable $align_with_prior_{ijk}$ as follows: $align_with_prior_{ijk} = 1$, if the sign of $Prior_{ijk}$ matches the sign of $Post_{ijk}$ (i.e., both are either positive or negative), and $align_with_prior_{ijk} = 0$, otherwise, when the signs do not match.

We define $n_align_with_prior_{ik} = \sum_{j=1}^5 align_with_prior_{ijk}$. Figure 8 shows the distribution of $n_align_with_prior_{ik}$ for participants whose initial decision in stage I is incorrect and see correct AI decisions in stage II. The distribution of $n_align_with_prior_{ik}$ for

participants in the AI condition is concentrated at 5, indicating that participants did not change prior beliefs across all five features. The distributions for the XAI and XAI+Reflection conditions are similar and more dispersed, indicating that they adjust beliefs in Stage II. The proportion of participants with $n_align_with_prior_{ik} = 5$ in the AI condition ($M = 0.63, SD = 0.07$) was significantly higher than that in the XAI condition ($M = 0.10, SD = 0.04, p < 0.001$) and XAI+Reflection condition ($M = 0.20, SD = 0.05, p < 0.001$). There were no significant differences between the XAI condition and XAI+Reflection conditions ($p = 0.316$). Thus, providing AI-driven decisions alone is not enough for participants to change their beliefs and finally accept AI suggestions. Reflection intervention encourages people to identify potential errors in AI outputs and their own biases, thereby enabling them to make more accurate decisions.

5.4 Analysis of Reflective Responses

After examining the effects of reflection on over-reliance, decision accuracy, and belief change, we conducted a qualitative analysis of the content of participants' reflections to address Research Question 4. We focused on three aspects: What types of conflict can people identify when prompted to reflect? What strategies or patterns do people use when making final decisions? Do overreliant or underreliant participants adopt different strategies for final decision making?

We analyzed the reflection texts from the XAI+Reflection group. Of 70 participants, 11 were excluded for providing irrelevant or trivial responses such as "I don't know". The final dataset consisted of valid reflections from 59 participants. The thematic analysis results are summarized in Figure 9, and the detailed findings are presented below.

5.4.1 Types of conflict identified between human and AI perspectives.

Analysis of participants' reflections revealed four distinct types of conflict that people were able to recognize when prompted to compare AI-provided information with their own initial reasoning:

① Conflict between initial decisions and AI recommendations. This theme captures moments when participants explicitly perceived a mismatch between their initial judgments and the AI's

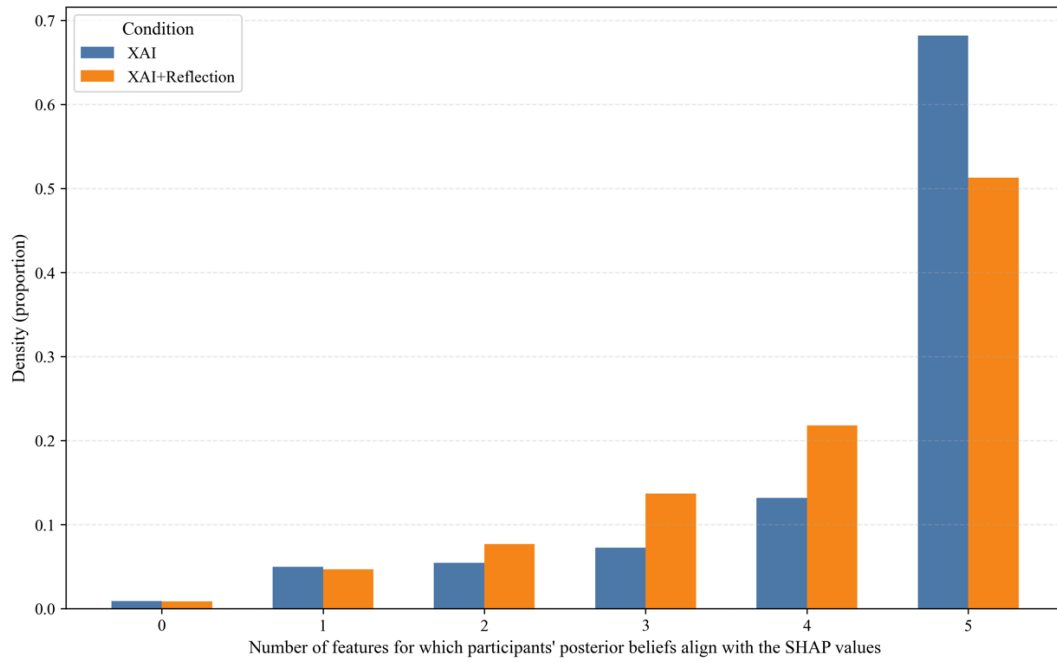


Figure 7: Number of features for which participants' posterior beliefs align with the SHAP values

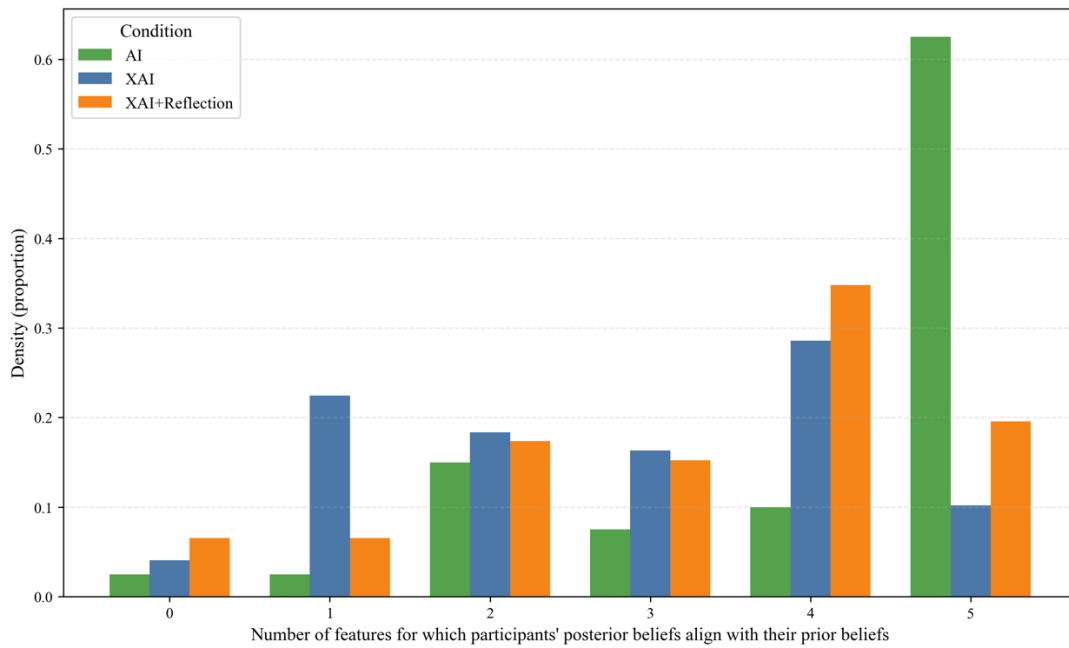


Figure 8: Number of features for which participants' posterior beliefs align with their prior beliefs

suggested outcome. For example, one participant noted that “the AI’s prediction does not match my initial judgment” (P4), indicating a direct recognition of discrepancy between human and AI decisions.

② Conflict between explanations and AI recommendations. Participants also identified contradictions within the AI system itself—specifically, inconsistency between the explanation and the model’s final prediction. For instance, one participant pointed out that

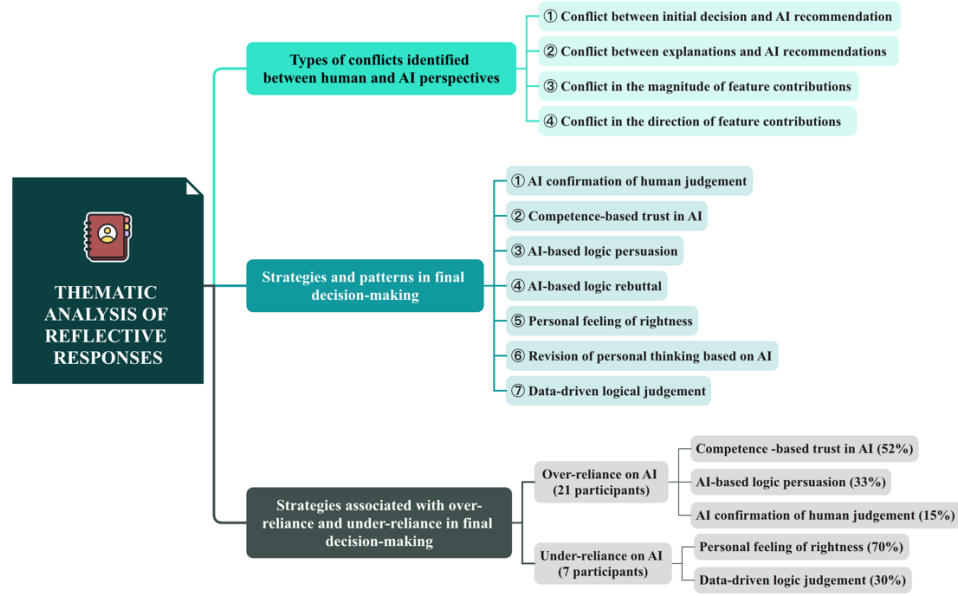


Figure 9: Thematic analysis of participants' reflection content.

“fasting plasma glucose has a 0.35 positive contribution, yet the AI still predicts no diabetes” (P5), while another remarked that “fasting plasma glucose suggests diabetes, but the AI concludes “no” because of insulin” (P43).

③Conflict in the magnitude of feature contributions. This theme refers to participants recognizing discrepancies between their own assumptions and the AI’s assessment regarding how strongly a feature contributes to diabetes risk. One participant stated, “I didn’t expect the diabetes pedigree function to matter so much” (P24), while another remarked, “I’m surprised diabetes pedigree function has such a high contribution” (P48). Similarly, the unexpectedly large weight of insulin level led one participant to note, “I’m surprised insulin has such a high weight” (P15).

④Conflict in the direction of feature contributions. Participants also noticed situations where the AI’s evaluation of a feature’s positive or negative influence contradicted their own expectations. For example, one participant emphasized that “BMI=27.7 should contribute positively to diabetes risk” (P18), whereas another expressed surprise that “BMI=27.7 actually lowers diabetes risk according to the AI” (P24).

5.4.2 Strategies and patterns in final decision-making. Analysis of participants’ final-decision explanations revealed seven distinct strategies that participants relied on to determine their final choice.

①AI confirmation of human judgment. Participants adopting this strategy made their final decisions because the AI’s analysis aligned with their initial judgments, which reinforced their confidence in those decisions. The key rationale is that AI validation strengthened their belief that their initial judgments were correct. For instance, one participant explained, “My judgment is consistent with the AI’s” (P15), while another noted, “I stick to my initial decision because my evaluations of each feature’s contribution

to diabetes risk are consistent with the AI’s conclusions, with no conflict” (P45).

②Competence-based trust in AI. This strategy reflects cases in which participants chose to follow the AI’s recommendation because they viewed the AI as more competent, systematic, or scientifically grounded than themselves. Their final choice was motivated by perceived superiority of AI expertise. Participants expressed sentiments such as “The AI is more scientific” (P12) and “The AI’s analysis is more comprehensive” (P35).

③AI-based logical persuasion. Participants who followed this strategy revised their final decision because they found the AI’s explanation logically convincing. Their choice was driven by the coherence of the AI’s reasoning. For example, one participant reasoned that “high fasting plasma glucose alone does not necessarily imply diabetes” (P11), leading them to adopt the AI’s “no-diabetes” output. Another explained that “the effect of insulin essentially offsets fasting plasma glucose” (P55).

④AI-based logical rebuttal. In contrast, some participants arrived at their final decision by actively identifying problems or inconsistencies in the AI’s reasoning. Their final choice, to stick with their initial assessment, was motivated by the conviction that the AI was flawed or unreliable in this instance. For example, participants commented, “I don’t trust the AI” (P7) and “The AI is inaccurate” (P57).

⑤Personal feeling of rightness. Under this strategy, participants made their final decision primarily based on confidence in their own intuition, knowledge, or diagnostic ability. Their reasoning centered on self-assurance rather than on any comparison between human and AI judgments. Participants expressed strong confidence, such as “I am fully confident in my ability to evaluate diabetes risk” (P13; similar statements by P36, P62).

⑥Revision of personal thinking based on AI. Some participants made their final decision by integrating insights from the AI into their own reasoning. Their final choice reflected adjustments to their initial judgment after reconsidering the AI's explanations. For example, one participant noted, "The AI made me realize the insulin level is indeed high" (P34), and another reflected, "I now understand that higher insulin is not always better" (P60).

⑦Data-driven logical judgment. For participants using this strategy, the final decision was primarily based on logical reasoning and numerical indicators rather than trust or distrust in either human or AI judgment. Their reasoning emphasized the use of objective thresholds or clinical benchmarks as the basis for final choice. For example, one participant stated, "152 mg/dL far exceeds the diagnostic threshold; this alone indicates diabetes" (P35), while another noted that "high fasting plasma glucose, insulin, and BMI all indicate high risk" (P22). These participants relied on data interpretation as the primary guide for their final decision-making.

5.4.3 Strategies associated with over-reliance and under-reliance in final decision-making. We further examined the strategies adopted by participants who exhibited over-reliance on AI or under-reliance on AI in their final decisions. The analysis reveals clear patterns in how these two groups reasoned.

Among the 21 participants who demonstrated over-reliance on AI, 52% relied on a competence-based trust in AI strategy, and 33% followed an AI-based logical persuasion strategy during final decision-making. These patterns suggest that overreliant participants tended to assume that the AI possessed superior capability, or they actively sought to rationalize the AI's recommendation. In doing so, they focused on justifying the AI's perspective rather than critically examining their own initial judgment.

In contrast, among the 7 participants who demonstrated under-reliance on AI, 70% relied on a personal feeling of rightness strategy, while 30% used a data-driven logical judgment strategy. These participants anchored their final decisions in personal confidence or selective data interpretation, leading them to disregard potentially useful AI feedback and persist in their own incorrect conclusions.

Taken together, these results indicate that individuals prone to over-reliance place excessive weight on AI-generated information and do not sufficiently verify their own reasoning, whereas those prone to under-reliance dismiss AI input without adequate consideration. Each strategy can lead to biased or suboptimal decisions. This highlights the need for tailored interventions that guide users with different tendencies toward more balanced and reflective decision-making.

5.5 The effect of reflection on user experience

For Research Question 5, Figure 10 summarizes participants' user experience across conditions. Specifically, perceived understanding of AI was significantly higher in the XAI+Reflection ($M = 5.76$, $SD = 0.09$) and XAI ($M = 5.73$, $SD = 0.09$) conditions compared to the AI condition ($M = 5.04$, $SD = 0.14$, $p < 0.001$). No significant difference was observed between the two XAI conditions ($p = 0.299$). This indicates that the XAI manipulation was successful. Providing SHAP explanations helped participants better understand how the algorithm arrived at its results.

Regarding trust in AI, participants in the XAI+Reflection ($M = 5.44$, $SD = 0.12$) and XAI ($M = 5.54$, $SD = 0.10$) conditions reported slightly higher trust compared to the AI condition ($M = 5.29$, $SD = 0.13$), though these differences were not significant.

Regarding cognitive load, participants in the XAI+Reflection condition reported higher mental demand ($M = 5.57$, $SD = 0.17$) than those in the XAI condition ($M = 5.37$, $SD = 0.19$), while the AI condition ($M = 5.48$, $SD = 0.18$) was in between. Complexity ratings showed a similar trend, with all conditions near the middle of the scale ($p > 0.05$). These results align with participants' responses during reflection, where they said that reflection promoted deeper thinking, resulting in higher perceived mental demand and complexity.

Finally, satisfaction did not differ significantly between the two XAI conditions (XAI: $M = 5.91$, $SD = 0.11$; XAI+Reflection: $M = 5.83$, $SD = 0.13$). However, both XAI conditions yielded higher satisfaction than the AI condition ($M = 5.56$, $SD = 0.15$), suggesting that the reflection manipulation did not significantly impact satisfaction.

Taken together, these results suggest that while reflection may modestly elevate perceived cognitive effort and reduce trust, these changes are not statistically significant. At the same time, reflection does not undermine satisfaction, and overall user experience remains more positive in XAI settings compared to AI baselines.

6 discussion

Our study demonstrated that guided reflection improves decision accuracy and reduces over-reliance on AI in AI-assisted decision-making. These benefits were especially pronounced among individuals with a high need for cognition and a high perceived understanding of AI. By analyzing participants' decision-making processes, we found that while participants in the AI and XAI conditions either seldom adjusted their beliefs or adjusted them uncritically toward the provided explanations, reflection prompted active, reasoned belief updating, ultimately enhancing decision performance. Furthermore, we identified five distinct types of conflicts participants encountered between their own reasoning and AI reasoning, and seven strategies they employed in their final decisions.

6.1 Conflict detection and decoupling as core mechanisms for prompting analytical engagement in AI-assisted decision making

Grounded in three-stage dual-process theory [41], our reflection intervention incorporated conflict detection and cognitive decoupling to encourage Type 2 processing during human-AI collaboration. The results show that reflection on targeted dimensions improves decision accuracy and reduces AI over-reliance, suggesting that reflection is a promising approach to mitigating cognitive biases in human-AI collaboration, such as anchoring on AI outputs [39] and confirmation bias [4].

Our results show that conflict detection and decoupling play important roles in fostering systematic thinking, consistent with Jussupow et al. [21]'s explanation of the human decision-making process in the context of human-AI collaboration. As Jussupow et al. [21] illustrated, detecting conflict triggers different forms

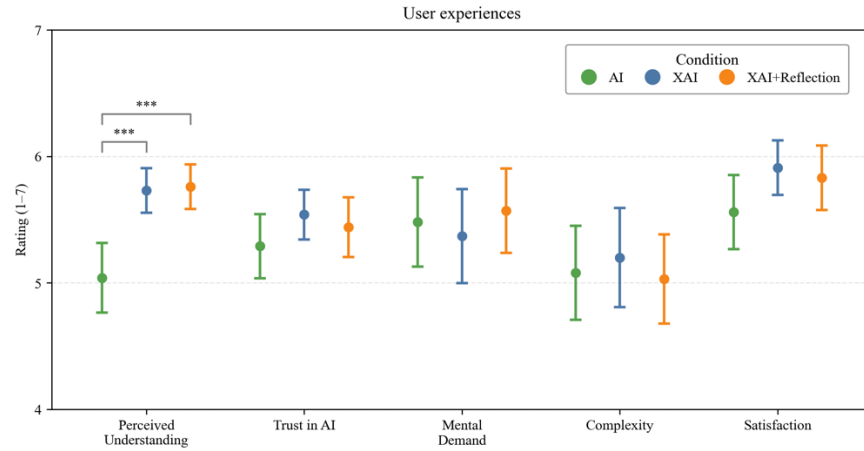


Figure 10: Participants' user experiences in different conditions. (*: $p < 0.001$)**

of metacognition and decision-making patterns, which in turn influence final decisions. This view resonates with the three-stage dual-process model [41], which posits that conflict detection is an early mechanism of Type 2 processing. Furthermore, Jussupow et al. emphasize that mere self-monitoring or system-monitoring does not eliminate biased decisions, aligning with the three-stage dual-process model's assertion that only decoupling allows individuals to override intuitive responses and correct biased judgments.

Beyond supporting existing theoretical accounts, our study also identifies key theoretical underpinnings for existing intervention strategies [7, 8, 12, 31, 40] aimed at addressing cognitive challenges in AI-assisted decision-making. We highlight the limitations of interaction-level designs [7, 40] in prompting individuals to detect conflict between AI and human judgment, while also recognizing that ideas such as “think the opposite [33]” and “deliberative AI [31]” provide valuable practical applications of conflict detection principles. Additionally, the “contrastive explanation framework [8],” which emphasizes presenting AI-generated explanations from both AI and human perspectives, supports the process of cognitive decoupling. These innovative directions collectively advance the field of AI-assisted decision making.

Our study contributes to advancing AI-assisted decision-making by empirically demonstrating the effectiveness of conflict detection and decoupling within a reflection intervention in practice. We hope that these two mechanisms will be incorporated into real-world human-AI collaboration, including interaction design, explanation design, and AI literacy training, to fully realize the synergistic potential of human-AI teams.

6.2 The role of need for cognition and perceived understanding of AI

The effectiveness of reflection varied across individuals. Consistent with prior research [7], and further validated in our study, reflection yielded greater benefits for participants with a high need for cognition. This suggests that cognitive interventions may be less effective for individuals low in need for cognition, an equity concern previously raised in the literature [1, 7]. Designing tailored

interventions that engage individuals with low need for cognition in analytic thinking will be essential to support more inclusive human-AI collaboration.

We also identified a moderating role of perceived understanding of AI, which captures individuals' mental models of AI systems [14]. While prior research has focused mainly on how explanation design [4] or social information [14] shapes perceived understanding of AI, our findings extend this line of work by showing that perceived understanding of AI moderates the impact of reflection on human-AI collaboration performance. This occurs because perceived understanding of AI influences conflict detection: when participants could interpret AI outputs and recognize potential conflicts, reflection facilitated appropriate belief updating; when such understanding was weak, reflection had a limited effect. Our findings underscore the importance of AI literacy initiatives designed to improve users' mental models of AI systems, as well as interaction designs that present AI reasoning in more interpretable forms.

6.3 Understanding human behavior in AI-assisted decision-making through the lens of belief change

Our analyses reveal that in the XAI condition, SHAP explanations influenced participants' beliefs across all diagnostic features in diabetes classification, even when participants doubted the plausibility of certain SHAP values (e.g., for BMI or age). In contrast, in the AI condition, participants rarely updated their beliefs about diagnostic features. This indicates that in AI-only settings, limited information restricts belief updates, causing individuals to rely on intuitive judgments or AI suggestions [21, 29, 41]. In XAI settings, explanations may inadvertently foster overconfidence in the algorithm, promoting heuristic reliance on AI competence rather than analytical evaluation [39]. These observations have important implications by explaining over-reliance and under-reliance through the lens of belief change [4]. Reflection reshapes participants' cognitive processes by discouraging blind acceptance of AI advice or rigid adherence to initial judgments. Instead, reflection encourages selective integration of AI suggestions with one's own prior beliefs. Our

study indicates that heuristic reasoning primarily drives both AI and XAI collaborations, and that reflection can steer belief updating toward more systematic reasoning.

6.4 Personalized reflection interventions and approaches for reducing cognitive burden

Through qualitative analysis of conflict types and final decision-making strategies, we found that reflection effectively enabled conflict detection, but the strategies employed thereafter were diverse and complex. Once individuals recognized discrepancies between their reasoning and AI outputs, they engaged in various information processing mechanisms, such as validating personal reasoning, scrutinizing system reasoning, learning from AI, or returning to the data itself. Our analysis of the decision-making strategies used by overreliant and underreliant individuals revealed that many still led to biased outcomes. This suggests that individuals with different predispositions (e.g., those who naturally trust AI [28] vs. those who rely heavily on themselves [43]) may benefit from personalized reflection interventions.

Although mental demand, perceived complexity, and satisfaction did not significantly differ between the XAI and XAI+reflection conditions, qualitative feedback indicated that reflection introduced additional cognitive burden into the interaction process. We suggest two directions for reducing this burden. On the interaction side, reflection activities could be gamified to make conflict detection and decoupling more engaging, and systems could further reduce user effort by supporting voice input or providing automated prompts to guide reflection. On the system side, improving the interpretability and clarity of AI outputs can lower the cognitive effort required to understand AI reasoning [50], enabling users to more easily validate AI decisions and learn from AI reasoning.

6.5 Limitations and future research

This study has several limitations that point to future research directions. First, we adopted SHAP as the explanation method, which is a feature-based, widely used, and state-of-the-art approach. Although SHAP has clear advantages, future studies could explore the effects of reflection when paired with different explanation types [24]. Individuals may have varying cognitive preferences for explanation formats, and tailoring explanation types to different users could enhance their ability to learn from AI, better understand how AI generates its predictions, and thereby achieve stronger reflection outcomes.

Second, the study employed diabetes prediction as the experimental task. While this domain is meaningful and relevant to human-AI decision making, the task itself is relatively less complex, and the accuracy gap between humans and the AI model was not large. As a result, the improvements in decision accuracy were limited in magnitude. Future research could examine more complex and unfamiliar tasks for lay participants, such as bird species identification [38] or medical image classification [17]. In these contexts, reflection may play a more critical role in enabling individuals to effectively integrate their own knowledge with AI-generated insights, potentially leading to more substantial improvements in performance.

Third, our measurement of cognitive processes relied primarily on self-reported and behavioral data, which offer limited resolution into real-time cognitive engagement and mental effort. Future studies could incorporate multimodal sensing techniques—such as eye tracking, electroencephalography (EEG), or functional magnetic resonance imaging (fMRI)—to capture fine-grained, physiological correlates of cognitive load, attention allocation, and belief updating during reflection. Combining such objective measures with reflective protocols would enable a more precise and holistic understanding of how guided reflection shapes cognitive processing in human-AI collaboration, while also helping to identify markers of effective versus superficial engagement.

7 Conclusion

In this paper, we examined reflection as a cognitive intervention in AI-assisted decision making. We conceptualized reflection as a structured process that encourages users to detect conflicts, reconsider their reasoning, and justify their final choices, thereby moving beyond purely technical approaches such as transparency or uncertainty visualization. We implemented a reflection intervention within an explainable AI (XAI) system and conducted a controlled experiment using a diabetes prediction task. Participants completed decisions across three conditions—AI, XAI, and XAI+Reflection. Our findings show that reflection improves decision accuracy and reduces over-reliance on AI. The benefits of reflection were particularly salient among individuals with a high need for cognition and a high perceived understanding of AI, underscoring the importance of individual differences in shaping the outcomes of cognitive interventions. Moreover, process-level analyses revealed that reflection guided participants to adjust their beliefs in a deliberate and rational manner, rather than rigidly adhering to initial judgments or blindly deferring to AI advice. These insights advance research on human-AI interaction in three key ways. First, this study demonstrated that reflection can serve as a simple yet powerful strategy to mitigate cognitive biases in AI-assisted contexts. Second, this study highlights the moderating roles of need for cognition and perceived understanding of AI, pointing to the need for personalized support in AI-assisted systems. Third, this study offers a belief-change perspective on over-reliance and under-reliance, providing a more fine-grained understanding of decision-making mechanisms in human-AI collaboration.

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A APPENDICES

A.1 Robustness check

To enhance the robustness of the findings, a multiple linear regression analysis was conducted to examine the effect of the reflection on the change in decision accuracy from Stage I to Stage II, while controlling for key covariates. The dependent variable, which represented the core construct of this study, was the change in decision accuracy from Stage I to Stage II. The model incorporated a set of predictors, including decision accuracy in Stage I (baseline control), a dummy variable for experimental group assignment ($0 = \text{XAI}$, $1 = \text{XAI} + \text{Reflection}$), AI-related background, need for cognition, and diabetes health literacy. The model evaluated whether the reflection component produced a unique and significant effect on the outcome after accounting for these factors.

The results of regression analysis are presented in Table A1. For the model predicting change in decision accuracy, the coefficient for the experimental group was positive and statistically significant ($\beta = 0.145$, $p = 0.029$). This indicates that participants in the XAI+Reflection group showed a significantly greater improvement in decision accuracy compared to the XAI group, after controlling for other variables.

Notes. $R^2 = 0.518$, Adjusted $R^2 = 0.497$.

Table 3: Results of multiple linear regression analysis predicting change in decision accuracy

	Dependent variable: change in decision accuracy		
	Beta	t-value	p-value
Experimental group (1=XAI+Reflection)	0.145	2.214	0.029
Decision accuracy in Stage I	-0.714	-10.913	<.001
AI-related background	0.004	0.060	0.952
Need for cognition	0.031	0.442	0.659
Diabetes health literacy	-0.063	-0.929	0.355